

Raspberry Pi OS Lite — Migration Checklist

Target: Raspberry Pi OS Lite (64-bit, Bookworm), headless, no desktop environment.
mpv renders directly via DRM/KMS on tty1.

Flash & First Boot

- Flash **Raspberry Pi OS Lite (64-bit, Bookworm)** via Raspberry Pi Imager
 - In Imager advanced settings: set hostname (eg. **tvPlayer**), enable SSH, configure WiFi, create user (e.g. **dp**)
 - Boot, SSH in
-

User & Groups

```
sudo usermod -aG video,input,gpio,render,audio dp
```

Autologin on tty1

Required for DRM/KMS access without a display manager.

```
sudo raspi-config  
# System Options → Auto Login → Console Autologin: Enable
```

Install Packages

```
sudo apt update && sudo apt install -y \  
  mpv libmpv-dev \  
  python3-pip \  
  git \  
  ffmpeg imagemagick \  
  mosquitto mosquitto-clients \  
  udisks2 lsof
```

Enable WebSocket listener for the web remote (browsers can't use raw MQTT).

Declaring **listener 1883** is required — Mosquitto 2.0+ disables the default port when any **listener** is added:

```
sudo tee /etc/mosquitto/conf.d/websockets.conf >/dev/null <<'EOF'
listener 1883
listener 9001
protocol websockets
allow_anonymous true
EOF

sudo systemctl restart mosquitto
```

USB Automount

OS Lite has no file manager to trigger USB mounting. **udisks2** handles it, but needs a polkit rule to allow mounting without an active desktop session:

```
sudo tee /etc/polkit-1/rules.d/10-udisks2-mount.rules >/dev/null <<'EOF'
polkit.addRule(function(action, subject) {
    if ((action.id == "org.freedesktop.udisks2.filesystem-mount" ||
        action.id == "org.freedesktop.udisks2.filesystem-mount-other-
seat" ||
        action.id == "org.freedesktop.udisks2.filesystem-mount-system")
    &&
        subject.user == "<YOUR_USERNAME>") {
        return polkit.Result.YES;
    }
});
EOF
```

Optionals

Hide kernel messages

Edit:

```
sudo nano /boot/firmware/cmdline.txt
```

Append (all on one line):

```
quiet loglevel=0 systemd.show_status=0 rd.systemd.show_status=0
vt.global_cursor_default=0 logo.nologo splash
```

Change CLI font size

```
sudo dpkg-reconfigure console-setup
```

Choose

- UTF-8
- Guess optimal character set
- Choose font **Terminus** and size to **16x32** (for fuzzy or TVs)

Custom splash image

```
sudo apt install plymouth plymouth-themes
```

And see here for installation: [assets/splashscreen/README.md](#)

Clone repo

```
git clone https://github.com/falue/tvPlayer.git
```

Python Dependencies

```
pip install --break-system-packages python-mpv evdev natsort paho-mqtt  
flask RPi.GPIO Pillow
```

`--break-system-packages` is required on Bookworm+ (PEP 668). Safe here since this is a single-purpose device.

Network / Hotspot (RaspAP)

Install RaspAP:

```
curl -sL https://install.raspap.com | bash
```

During the installer, choose:

- **Complete installation** — yes
- **Set default config for hostapd** — yes
- **Enable HttpOnly for session cookies** — yes
- **Enable control service** — yes

- **Install Ad-blocking** — no
- **Install OpenVPN** — no
- **Install WireGuard** — no
- **Install RestAPI** — no
- **Enable VPN provider client** — no
- **Enable TCP BBR congestion control** — yes

After reboot, RaspAP creates a WiFi AP with default SSID **raspi-webgui**.

Fix: NetworkManager vs hostapd conflict

If **wlan0** stays in **type managed** instead of **type AP**, NetworkManager is fighting hostapd for the interface. This happens when a saved Wi-Fi client profile auto-reconnects.

1. Disable autoconnect of the old Wi-Fi client profile:

```
sudo nmcli connection modify netplan-wlan0-<SSID> connection.autoconnect
no
sudo nmcli connection down netplan-wlan0-<SSID>
```

2. Tell NetworkManager to completely ignore wlan0:

```
sudo mkdir -p /etc/NetworkManager/conf.d

sudo tee /etc/NetworkManager/conf.d/10-unmanaged-wlan0.conf >/dev/null
<<'EOF'
[keyfile]
unmanaged-devices=interface-name:wlan0
EOF

sudo systemctl restart NetworkManager
```

3. Reset the interface and restart hostapd (one-time, not needed after reboot):

```
sudo ip link set wlan0 down
sudo ip addr flush dev wlan0
sudo ip link set wlan0 up
sudo systemctl restart hostapd
```

4. Verify:

```
iw dev
```

Expected: `type AP` and `ssid RaspAP` (not `type managed`).

After reboot, hostapd should automatically bring up `wlan0` as AP with IP `10.3.141.1`. Verify with a reboot.

Configure the RaspAP

Configure via the RaspAP admin panel at `http://10.3.141.1` from another device.

RaspAP admin panel login default: user `admin`, password `secret` — change pw to eg `digitalProps#8400`

- **Hotspot → Basic:** SSID = `tvPlayer`, Security = WPA2
- **Hotspot → Security:** set a password
- **DHCP Server:** leave defaults (clients get `10.3.141.x`)

The tvPlayer web remote will then be available at `http://10.3.141.1:8080` from any device connected to the `tvPlayer` WiFi.

Systemd Service

```
sudo nano /etc/systemd/system/tvplayer.service
```

```
[Unit]
Description=tvPlayer
After=multi-user.target
StartLimitIntervalSec=30
StartLimitBurst=10

[Service]
User=dp
WorkingDirectory=/home/dp/tvPlayer
Environment=XDG_RUNTIME_DIR=/run/user/1000
ExecStartPre=/usr/bin/systemctl --user restart pipewire pipewire-pulse
wireplumber
ExecStartPre=/usr/bin/sleep 1
ExecStart=/usr/bin/python3 -u tvPlayer.py
Restart=on-failure
RestartSec=2
StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target
```

`Environment=XDG_RUNTIME_DIR=/run/user/1000` is **not optional** — it is how mpv finds the user PipeWire socket. Check the real uid with `id -u dp` first. Without it mpv cannot reach PipeWire and there is no audio at all. See the *Audio Setup* section below.

Enable:

```
sudo systemctl enable tvplayer.service
```

Verify DRM Works

```
mpv --vo=drm --drm-connector=HDMI-A-1 /path/to/test.mp4
```

If this plays fullscreen without X, the base system is ready.

Refactor Steps (after OS is ready)

1. Replace pygame + socat IPC → python-mpv
2. Disable all mpv built-in UI (osc, osd, keybinds)
3. Replace pygame keyboard → evdev thread
4. Configure `vo=drm`, systemd service on tty1
5. Multi-HDMI hotplug (EDID poll, connector select)
6. Overlay routing via python-mpv
7. Multi-monitor: one process, multiple players, per-screen state

Audio Setup — Sound Out of Every Output at Once

Goal: audio plays simultaneously from the **3.5mm analog jack** and **both HDMI ports**, so you can plug into whichever you like and it just works.

By default mpv outputs to a single device. The fix is a PipeWire **combined sink**: one virtual output that mirrors audio to every real output present on the system.

Volume and mute stay in tvPlayer. mpv applies them in software (`player.volume`) *before* the audio reaches the combined sink, so whatever you set in the webremote applies to all outputs equally. Do not use `pactl set-sink-volume` or `alsamixer` to control level — leave those at 100% and let the webremote own it. Otherwise you end up with two independent volume stages fighting each other.

Prerequisites

```
sudo apt install pipewire pipewire-pulse wireplumber
sudo apt install pulseaudio-utils
```

Verify PipeWire is running:

```
pactl info | grep "Server Name"
# Should show: PulseAudio (on PipeWire ...)
```

Create a Combined Sink

Create the config file:

```
mkdir -p ~/.config/pipewire/pipewire.conf.d
nano ~/.config/pipewire/pipewire.conf.d/combined-sink.conf
```

Paste:

```
context.modules = [
  {
    name = libpipewire-module-combine-stream
    args = {
      combine.mode = sink
      node.name = "combined-output"
      node.description = "HDMI + Analog Combined"
      combine.latency-compensate = true

      stream.rules = [
        {
          matches = [
            { media.class = "Audio/Sink" node.name =
"~alsa_output.*" }
            { media.class = "Audio/Sink" node.name = "pi-
hdmi-0" }
            { media.class = "Audio/Sink" node.name = "pi-
hdmi-1" }
          ]
          actions = { create-stream = {} }
        }
      ]
    }
  }
]
```

This mirrors audio to **all** available sinks. The `matches` rule deliberately targets every `Audio/Sink` rather than naming devices, which matters for two reasons:

- On a Pi 4 the outputs appear as `vc4-hdmi0`, `vc4-hdmi1` and the analog `bcm2835-headphones` — naming them individually is brittle.
- **An HDMI sink only exists while a display is plugged in.** The `combine-stream` module picks up sinks as they appear and drops them as they vanish, so hotplugging a second screen adds its audio automatically with no reconfiguration.

Set as Default

Restart PipeWire:

```
systemctl --user restart pipewire pipewire-pulse wireplumber
```

List sinks and find the combined one:

```
pactl list short sinks
```

Set it as default:

```
pactl set-default-sink combined-output
```

To make it persistent across reboots, add to `~/.config/pipewire/pipewire.conf.d/default-sink.conf`:

```
context.properties = {  
    default.audio.sink = "combined-output"  
}
```

Make it work under systemd

PipeWire runs as a **user** service, but `tvPlayer` is started by a system unit. Two things are needed so the player can reach the user's audio session:

1. Keep the user session alive even when nobody is logged in:


```
sudo loginctl enable-linger dp
```

2. Confirm the service runs as that same user and can see the session bus. The unit file lives **on the Pi** at `/etc/systemd/system/tvplayer.service` (see the *Systemd Service* section of [SETUP_LITE.md](#) for how it is created — the `tvplayer.service` file in this repo is only a reference copy, systemd does not read it):

```
sudo nano /etc/systemd/system/tvplayer.service
```

It already contains `User=dp` in its `[Service]` block. If audio is silent when started by systemd but works when you run `python3 tvPlayer.py` by hand, add one line to that same `[Service]` block:

```
[Service]
User=dp
WorkingDirectory=/home/dp/tvPlayer
ExecStart=/usr/bin/python3 -u tvPlayer.py
Environment=XDG_RUNTIME_DIR=/run/user/1000
...
```

Check the real uid first with `id -u dp` — it is usually `1000`. Then reload and restart:

```
sudo systemctl daemon-reload
sudo systemctl restart tvplayer
```

Verify

List sinks and confirm the outputs are all there:

```
pactl list short sinks
```

Test the combined sink directly:

```
speaker-test -c 2 -t wav
```

You should hear audio from the analog jack and every connected HDMI display at once.

Then confirm mpv is actually feeding the combined sink while tvPlayer runs:

```
pactl list short sink-inputs
```

Notes

- Only the outputs that physically exist produce sound. With one HDMI connected you get that HDMI plus analog; plug in the second and it joins automatically.
- mpv needs no special audio configuration — it follows the system default sink.
- `hdmi_drive=2` in `/boot/firmware/config.txt` does **nothing** on Bookworm — it was a legacy-firmware setting and the `vc4-kms-v3d` driver ignores it. Do not go looking there.
- `combine.latency-compensate = true` aligns the outputs. If you hear echo from having both a TV speaker and a jack-connected speaker in the same room, that is the physical distance between them, not a config problem.
- If PipeWire is unavailable, PulseAudio's `module-combine-sink` achieves the same thing.

Status

Untested. Expect to iterate on this. `journalctl --user -u pipewire -n 50` is the first place to look when something does not come up.